



December 2025

EE2211-1 – ROBOTICS AND AUTOMATION

(For EE)

Duration: 3 Hours

Max. Marks:100

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Fivefull** questions choosing **Two full** questions from **Unit – I &Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. A simple joint used in robots for rotational movement only is called _____ joint.
A) prismatic B) spherical
C) revolute D) Cylindrical
2. The number of independent degrees of freedom for a manipulator is typically at least _____.
A) 3 B) 4
C) 5 D) 6
3. Homogeneous transformation in robotics consists of _____.
A) A 3x1 rotation vector and a 3x3 translation matrix B) A 3x3 rotation matrix and a 3x1 translation vector
C) 4x4 rotation matrix and a 4x1 translation vector D) 4x1 rotation vector and a 4x4 translation matrix
4. Which of the following statement is true with respect to the relationship between the reachable workspace and the dextrous workspace?
A) The dextrous workspace consists of those points that the manipulator can reach with an arbitrary orientation of the end-effector, and it is a subset of the reachable workspace
B) The reachable workspace is a subset of the dextrous workspace
C) The reachable workspace is for open-loop robots, and the dextrous workspace is for closed-loop robots
D) There is no difference; the terms are interchangeable
5. _____ type of robot power source is primarily used for lifting heavy loads due to its unrivaled speed of response and torque?
A) Pneumatically powered B) Hydraulically powered
C) Electrically powered (DC or AC servo motors) D) Solar powered
6. _____ sensor characteristic describes how close the output is to the true value.
A) Precision B) Reliability
C) Accuracy D) Speed of response
7. The technical term for the smallest increment of motion that the controller of a robot can sense is called _____.
A) controller repeatability B) controller reliability
C) controller accuracy D) controller resolution
8. A continuous path robot in terms of control is characterized by _____.
A) the entire path of the end-effector that can be controlled B) its movement limited to predetermined mechanical stops
C) the sophisticated controller that makes it a most advanced robot D) its ability to get taught by a discrete set of points; however, with no control on the path of the end effector in between taught points

9. _____ refers to the control of a robot along a continuous path over time.
 A) Set-point tracking B) Actuator dynamics
 C) Trajectory tracking D) Reliability
10. Inexpensive and simple, but lacking precise control, _____ type of robot power source has limited applications.
 A) electrically powered B) pneumatic
 C) hydraulic D) servo
11. _____ component within a magnetic sensor's chip is used to detect a magnetic vector.
 A) Photodetector B) Piezoelectric
 C) Laser diode D) Magnetoresistive
12. The hydraulic actuators work on the principle of _____.
 A) Newton's Laws of motion B) Archimedes' Law
 C) Pascal's Law D) Kirchhoff's Law
13. The process in robotics of calculating the necessary joint angles of a robot arm to move its gripper to a desired position and orientation is called _____.
 A) Inverse kinematics B) forward kinematics
 C) velocity kinematics D) acceleration kinematics
14. Image processing helps robots in _____.
 A) navigation only B) navigation and inspection only
 C) navigation and manipulation only D) navigation, inspection and manipulation
15. Denvit-Hartenberg representation is used to _____ in robots.
 A) formulate Euler-Lagrange equations B) standardize the coordinate frames for spatial linkages
 C) control the sensors and actuators D) navigate and manipulate the grippers
16. _____ in robotics is an iterative, two-phase algorithm used to compute the dynamics of a robot arm.
 A) Denvit-Hartenberg convention B) Euler-Lagrange equation formulation
 C) Newton-Euler formulation D) Forward kinematics
17. _____ configuration of robots combines both linear and rotary joints to achieve a polar work envelope.
 A) Spherical B) Cylindrical
 C) Cartesian D) SCARA (Selective Compliance Assembly Robot Arm)
18. The core language used in robots for high performance control and low-level hardware access is _____.
 A) COBOL B) JavaScript
 C) C# D) C++
19. Lead through programming is an example of _____ method for robots.
 A) offline programming B) online programming
 C) standalone programming D) special programming
20. _____ is the process of creating an automated workstation by integrating a robot with peripheral equipment to perform tasks like assembly, welding, or inspection
 A) Optimizing the robots B) Operating the robots
 C) Robots cell design D) Robots testing

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) With the appropriate block diagrams, briefly explain (i) the Articulated Configuration (RRR) and (ii) the Spherical Manipulator Configuration (RRP).	8	L*2	1	1
	b) List out the different types of proximity sensors based on their working principle and their applications in robotics.	8	L1	2	1
2.	a) Write a note on homogeneous transformation in Robotics.	10	L2	1	1

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b)	What are the key functions and applications of magnetic sensors in robotics?	6	L2	2	1
3. a)	What are the different criteria based on which the robots are classified? Explain.	8	L2	1	1
b)	Explain the steps involved in the designing of grippers.	8	L2	2	1

Unit – II

4. a)	Explain the concept of set-point tracking control in robotics.	6	L2	3	1
b)	Create the Denavit-Hartenberg parameter table by drawing the kinematic diagram of a two degrees of freedom of a robotic arm.	10	L3	4	1
5. a)	Explain the feed forward method of control in robots.	8	L2	3	1
b)	Explain the step-by-step approach based on Denavit-Hartenberg convention for deriving the forward kinematics for any manipulator.	8	L3	4	1,2
6. a)	What is velocity kinematics? Also, explain (i) forward velocity kinematics (ii) inverse velocity kinematics (iii) importance and limitations of velocity kinematics.	8	L2	3	1
b)	Write a note on Newton-Euler formulation in robotics.	8	L2	4	1

Unit – III

7. a)	Write a note on common programming languages used in robots programming.	10	L2	5	1
b)	Explain the key steps in the robots cell design.	6	L3	5	1,2
8. a)	Write a note on the robots programming methods.	8	L2	5	1
b)	Explain the applications of robots in assembly and inspection.	8	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO*, Course Outcome; PO* Program Outcome
